

# **BT601 Final Term File**

## **By All About Biotech**

### Objectives

1. Father of microbiology
2. First discovered term vaccine
3. Who discovered first cell line
4. Immunodiffusion is performed using -----%agarose
5. Tissue sample stored at degree for years
6. Toroviruse belong to a family
7. Biggest virus name
8. Virus infection cell release
9. Smallest viruse name
10. Toroviruse belong family
11. Virus infection cell release
12. Misfold if protein
13. Mad cow

### Subjective

#### 2 Marks Questions

**1. Define intron and exon**

Introns:- The non coding region on the viral genome.

Exons:- The coding regions on the viral genome.

**2. Define lysogens?**

A lysogen or lysogenic bacterium is a bacterial cell in which a phage exists as DNA in its dormant state (prophage)

**3. Viroid transmission?**

• By aphids • By cross contamination • Due to mechanical damage to plants as a result of horticultural or agricultural practices • From plant to plant by leaf contact

**4. Structure of prions?**

The protein that prions are made of (PrP) is found throughout the body, even in healthy people and animals. PrPC is a normal protein found on the membranes of cells. It has 209 amino acids (in humans) One disulfide bond, A molecular mass of 35–36 kDa A mainly alpha-helical structure. The normal protein is not sedimentable; meaning that it cannot be separated by centrifuging techniques.

#### **5. Fluorescence?**

The ability of substances to absorb short wavelengths of light (ultraviolet) and give off light at a longer wavelength(visible). It is used in fluorescence microscopy.

#### **6. T-phage?**

Head and tail • Called binal symmetry • Icosahederal head and helical tail symmetry.

#### **7. Berk filter ?**

- Manufactured of diatomaceous Earth
- Graded into three porosities
- The V (veil) grade is coarse and Used for preliminary filtration to remove bacteria or large particles from a solution
- The N (normal) grade is most useful since it retains most bacteria, while all viruses will pass through
- The W (Weing) grade has very small pores that will retain all bacteria and large viruses

#### **8. Picorna viruse?**

- A picornavirus is a virus belonging to the family Picornaviridae, and order Picornavirales
- Vertebrates, including humans, serve as natural hosts. Birds also serve as host
- Picornaviruses are nonenveloped viruses that represent a large family of small, cytoplasmic, plus-strand RNA (~7.5kb) viruses . • 30-nm icosahedral capsid.

#### **9. Name of plant which have positive RNA ?**

Bromovirus • Ilavirus • Alfamovirus • Ourmiavirus • Tobamovirus • Hordievirus • Pecluvirus • Furovirus • Pomovirus

#### **10. What is meant by Paramyox ?**

Para means beyond and myox means mucus.

**11. Phase growth curves steps?**

It consist of four steps

1. Inoculation
2. Eclipse
3. Burst
4. Burst size

**12. Magnification?**

It is the ratio of size of the an object seen under the microscope to the actual size observed with Unaided eye.

**13. Inclusions?**

Examination of virus infected cells by light or electron microscope shows viral multiplication and maturation referred as 'inclusions' or 'factories'.

**14. Write the two names of plant viruses?**

Tobacco mosaic virus and Potato virus.

**15. What is the polyribosomes?**

- Most cellular protein is assembled and synthesized on aggregates of ribosomes called polyribosomes

**16. Problem facing during the viral concentration?**

One infectious unit equals to  $10^3$  to  $10^5$  noninfectious particles

- Low ratio associated with slow uncoating of viral particles
- Exception of vaccinia virus

**17. Symmetry of the phage?**

18. Which media is used while transfer the viral infection?

**19. Principle of hemagglutination?**

**20. Cell culture?**

**21. Positive sense RNA virus of plant?**

**22. Viruse neutralization test**

**23. Latent occult of virus infection?**

**24. Mechanism of vaccine?**

**3 Marks Questions**

**1. Uncoating of viruses**

- Uncoating means removal of capsid
  - Exposing the viral genome in the host cell Host range of viruses
- The Protein coat around genome gets disintegrated Cytoplasm Cellular enzymes
- Makes genome ready to takeover the host machinery
  - Activates cascade of the steps required for viral replication

## 2. Types of mutant virus?

- Biochemical markers: drug-resistance mutations
- Deletions: similar to non-sense mutations
- Host range: refers to whole host animal
- Non-sense mutation: alteration of coding sequence
- Plaque morphology
- Temperature sensitive
- Cold sensitive
- Reverents

## 3. In vitro genetic technique used in virology?

- Physical analysis
- Isolation of nucleic acids from virus preparations
- Direct analysis of DNA using restriction endonucleases enzyme
- Use of vectors in DNA sequencing
- Electron microscopy
- RNA sequence analysis using Reverse transcriptase

## 4. Sub families of the herpesvirus ?

Alphaherpesvirinae • Latent infections in sensory ganglion, genome size 120-180kbp • Simplexvirus, Varicellovirus, Human herpes virus-I and II, • Human herpes virus-III

Betaherpesvirinae • Restricted host range, genome size 140-253 kbp • Cytomegalovirus, Muromegalovirus, Roseolovirus, Human herpes virus-V, Mouse cytomegalovirus-I, Human herpes virus-VI and VII Gammaherpesvirinae • Infection of lymphoblastic cells, genome size 105-175 kbp • Lymphocryptovirus, Rhadinovirus, Human herpesvirus-IV, Human herpes virus-VIII

## 5. Technique for the detective viral growth?

- Death of embryo
- Damage to embryo
- Pocks or lesions on egg membrane

## 6. LD50?

It stands for the lethal dose 50.

## 7. Replication of togavirus?

- Viral RNA acts as both genome as well as mRNA
- After fusion of viral capsid with cell receptor
- RNA released in cytoplasm

8. What is the effect of viral infection on host cell.

9. What is the purpose of strand amplification?

10. Target of the evasion protein?

11. Requirement to initiate the viral diseases?

12. Three evolution theories?

13. Red queen hypothesis?

14. Three filtration methods?

15.

## 5 Marks Questions

### 1. Membrane technology application?

Cold sterilization of beverages and pharmaceuticals

Clearing of fruit juice, wines and beer

Separation of bacteria from water

Separation of oil/ water emulsions

Reverse Osmosis

Solid-liquid separation for pharmacies or food industries

### 2. Factors involved in virus titration?

1. Incubation time:-

Precise period • Most viruses take precise time to reproduce • Time between inoculum and appearance of symptom.

2- Types of virus

Plant virus or animal virus • Conditions provided • Optimum media

3- Procedure contamination

Bacterial contamination, fungal contamination, careless lab work, Improper cleanliness and sanitization, Disruption of viral genome.

### 3. Structure of plant viruses?

These are extremely small and can only be observed under an electron microscope. Over 50% of known plant viruses are rod-shaped (flexuous or rigid).

- The length of the particle is normally dependent on the genome but it is usually between 300–500 nm with a diameter of 15–20 nm. The second most common structure amongst plant viruses are isometric particles.
- They are 25–50 nm in diameter.
- In cases when there is only a single coat protein, the basic structure consists of 60 T subunits, where T is an integer.
- Some viruses may have 2 coat proteins that associate to form an icosahedral shaped particle.

### 4. Corona viruses replication?

- Attachment of the viral S protein receptors
- Endocytosis of the virus into the host cell
- Fusion of virus membrane with endosomal membrane
- Genome is released into the cytoplasm.
- Synthesis and proteolytic cleavage of the replicase polyprotein.

A dsRNA genome is synthesized from the genomic ssRNA(+). • The dsRNA genome is transcribed by providing viral mRNAs/new ssRNA(+) genomes. • Synthesis of structural proteins encoded by subgenomic mRNAs. • Assembly and budding at membranes of the endoplasmic reticulum (ER), the intermediate compartments, and/or the Golgi complex. • Release of new virions by exocytosis.

### 5. Step of lytic cycle?

It has six steps

1. Attachment
2. Penetration
3. Transcription
4. Replication of phage DNA
5. Assembly
6. Release

### 6. How normal cell transform into cancerous cell?

Oncogene – a gene that when mutated or expressed at abnormally high levels contributes to converting a normal cell into a cancer cell

### 7. Write genome and structure of retrovirus?

Genome of retrovirus:-

- Single stranded RNA viruses
- Positive sense
- enveloped

- Changes to DNA by RT enzyme

There are three important gene

1. GAG
2. ENV
3. POL

Structure of retroviruses:-

It is the single stranded positive sense RNA viruses. It is Icosahedral in shape.

All the retrovirus are surrounded by viral envelope. It contain two identical RNA and reverse transcriptase enzyme.

## 8. Replication of retroviruses?

1. Early steps from the cell surface to the nucleus
  - Retroviral particles must bind specifically to their target cells • cross the plasma membrane • reverse-transcribe their RNA genome • Uncoat the cores • find their way to the nuclear membrane • penetrate into the nucleus and Hijack the cellular machinery

2- LATER STAGE

- Transcription
- Transduction
- Assembly
- Maturation
- Budding
- Release

9. Baltimore Classification?

10. Rotter tube cultures?

## 10 Marks Questions

1. How many ways in which viruses work to produce infection in human body ?
2. Working of viruse to affect the host cell?
3. What are microscope? Explain the types of microscope?
4. Viruses transmitted in plant method?
5. Centrifugation and types?